

1. Administrative & Study Identification

Full Study Title: A Study of Bomedemstat (MK-3543) in Participants With Mild or Moderate Hepatic Impairment (MK-3543-023) **Short Title/Acronym:** MK-3543-023 Hepatic Impairment Study **Protocol Number:** MK-3543-023 **NCT Number:** NCT07049939 **Sponsor Name:** Merck Sharp & Dohme LLC **Investigational Product:** Bomedemstat (MK-3543) **Phase of Development:** Phase 1 **Medical Monitor:** Merck Sharp & Dohme LLC Clinical Director

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the effect of mild or moderate hepatic impairment on the single-dose pharmacokinetics (PK) of bomedemstat (AUC0-inf, Cmax) and assess its overall safety and tolerability compared to healthy matched controls. **Secondary Objectives:** To determine additional pharmacokinetic parameters of bomedemstat, including AUC0-last, AUC0-24hr, C24, Tmax, t1/2, apparent clearance (CL/F), and apparent volume of distribution (Vz/F).

2.2 Background & Rationale Bomedemstat (MK-3543) is an investigational oral lysine-specific demethylase 1 (LSD1) inhibitor currently being developed for the treatment of myeloproliferative neoplasms, including essential thrombocythemia and myelofibrosis. Because liver impairment can significantly alter the metabolism and clearance of systemically administered drugs, this Phase 1 study is necessary to evaluate the pharmacokinetic profile and safety of bomedemstat in patients with chronic, stable liver insufficiency compared to healthy volunteers, helping to establish appropriate dosing guidelines for this population.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional (Pharmacokinetic Study) **Control Type:** Active Comparator (Healthy Matched Controls) **Blinding:** Open-label **Randomization:** N/A (Participants allocated to cohorts based on hepatic function: mild impairment, moderate impairment, or healthy matched controls)

3.2 Planned Enrollment & Duration Target \$N\$: 24 participants **Number of Sites:** Multiple clinical pharmacology sites in the US and globally (e.g., Orlando Clinical Research Center, The Texas Liver Institute, University of Miami) **Estimated Study Start:** Nov-2024 **Estimated Primary Completion:** Mar-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adult participants with chronic, stable mild or moderate hepatic impairment (based on Child-Pugh scale) or healthy matched controls. 2. Is a non-smoker or is a moderate smoker for at least 3 months prior to dosing. 3. Healthy controls must match the mean age (± 15 years), mean body-mass index (BMI) ($\pm 25\%$), and sex ratio (± 2) of the participants with mild and moderate hepatic impairment.

4.2 Exclusion Criteria 1. Severe complications of liver disease within the preceding 3 months, history of recent variceal bleeds, or evidence of hepatorenal syndrome. 2. History of liver or other solid organ transplantation, or has a transjugular intrahepatic portosystemic shunt (TIPS) / portacaval shunting. 3. Healthy controls: History of clinically significant endocrine, cardiovascular, hematological, hepatic, immunological, renal, respiratory, genitourinary, or major neurological diseases.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: A single 25 mg dose of bomedemstat administered on Day 1. **Route of Administration:** Oral **Duration of Treatment:** 1 day (single dose), followed by an intensive 216-hour pharmacokinetic sampling period and a 14-day safety follow-up.

5.2 Concomitant & Prohibited Therapy Permitted: Standard stable maintenance medications for baseline hepatic impairment not known to interfere with bomedemstat metabolism. **Prohibited:** Use of strong CYP inhibitors or inducers, regular use of cannabis products within 6 months of study, and active infections requiring systemic therapy.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Days -28 to -1	Informed Consent, Medical History, Child-Pugh Scoring, Matching Assessments
Baseline/Dosing	Day 1	Admission, Baseline Vitals, Single Oral Dose of Bomedemstat (25 mg)
PK Monitoring	Up to 216 hours post-dose	Intensive serial blood sampling for AUC and Cmax determination, Safety Monitoring
End of Study	Day 14	Final Safety Labs, Vital Signs, Exit Evaluation, AE Recording

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Pharmacokinetics of bomedemstat assessed via Area Under the Curve from time zero to infinity (\$AUC_{0-inf}) and Maximum Plasma Concentration (\$C_{max}).

Measurement Method: Serial blood sampling and central laboratory liquid chromatography-tandem mass spectrometry (LC-MS/MS) analysis.

7.2 Secondary & Exploratory Endpoints 1. Additional PK parameters: \$AUC_{0-last}, \$AUC_{0-24hr}, \$C₂₄, \$T_{max}, \$t_{1/2}, \$CL/F\$, and \$V_z/F\$. 2. Number of participants experiencing Adverse Events (AEs) and number of participants discontinuing study treatment due to an AE.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard vital signs, clinical laboratory evaluations, and physical examinations monitored from informed consent through the 14-day follow-up. **Serious Adverse Events (SAEs):** Required to be documented and reported to the sponsor and regulatory authorities within standard 24-hour timelines. **Data Monitoring Committee (DMC):** Internal safety evaluation handled by the sponsor (Merck Sharp & Dohme LLC).

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: * *Safety Set:* All participants who received at least one dose of bomedemstat. * *PK Population:* Participants with evaluable concentration-time profiles. **Sample Size Justification:** A target enrollment of \$N=24\$ provides adequate precision to estimate PK parameters in mild/moderate hepatic impairment compared to healthy controls using established FDA/EMA guidance for hepatic impairment studies. **Handling of Missing Data:** Standard non-compartmental analysis (NCA) procedures will be applied; missing values will be handled per the sponsor's standard operating procedures for Phase 1 PK trials.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Routine clinical site monitoring visits will be conducted to ensure protocol adherence, accurate data transcription, and safety compliance. **Audit Trail:** Validated 21 CFR Part 11 compliant Electronic Data Capture (EDC) system with comprehensive audit trails for all data entries.

11. Study Identifiers & Governance

Primary ID: MK-3543-023 **Registry Number:** NCT07049939 **Sponsor/Lead Organization:** Merck Sharp & Dohme LLC **Study Title:** MK-3543-023 Hepatic Impairment Study **Official Regulatory Title:** A Study of Bomedemstat (MK-3543) in Participants With Mild or Moderate Hepatic Impairment (MK-3543-023)

12. Clinical Framework (Hepatic Impairment Specific)

Primary Condition: Hepatic Impairment (Mild or Moderate)
Diagnosis Threshold: Child-Pugh score classification
Medical Methodology: Pharmacokinetic single-dose exposure evaluation
Optimization Target: Dosing characterization for future indications

13. Study Procedures & Methodology

Management Pathway: Clinical pharmacology Phase 1 inpatient/outpatient standard monitoring
Education Protocol (HCP): Investigator brochure review & Phase 1 unit procedural training
Education Protocol (Patient): Dosing restrictions, dietary requirements, and study timeline review during screening
Quality Audit Mechanism: Independent QA auditing of the Phase 1 clinical research facilities

14. Observation & Follow-Up Schedule

Total Duration: Approximately 4-6 weeks per participant (including screening)
Onsite Visit Cadence: Intensive monitoring spanning Day 1 to Day 9 (216 hours)
Remote Monitoring Frequency: N/A (Standard Phase 1 Unit observation)
Checklist Review Interval: Daily during confinement/visit windows

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	____	[DD-MMM-YYYY]				